JIAYU TANG

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PROFESSIONAL SUMMARY

I am interested in utilizing interdisciplinary approaches to study how antibodies bind antigens. I aim to use AI rigorously, grounded in and validated against experimental data, to build a mechanistic understanding of antibody –antigen interactions that can inform clinical and industrial decisions as well as frontier research.

EDUCATION

M.S., Food Science and Technology; Minor in Statistics **2024–2026**
University of Nebraska–Lincoln, Lincoln, NE
Awards: Excellence in the Department of Food Science; Mary and Charles C. Cooper/Emma I. Sharpless Fellowship | **Funding:** Chancellor's Graduate Research Grant; Graduate Research Assistantship

Dual Bachelor's Degree Program **2020–2024**
B.Sc., Food Science and Technology *University of Nebraska–Lincoln, Lincoln, NE*
B.Eng., Food Science and Engineering *Northwest A&F University, China*
Awards: Undergraduate Merit Scholarship

RESEARCH EXPERIENCE

OPS Research Assistant **10/2026–Present**
Department of Infectious Diseases and Immunology, University of Florida, Gainesville, FL
- Run deep-learning model (AlphaFold3, ESMFold2) and antibody design workflows (RFantibody, BindCraft2, RFD4) for antibody-antigen predictions and evaluate model performance.
- Use AI coding agents (Claude Code, Codex) to plan, implement, and audit in-silico antibody-design pipelines built on framework grafting, native-structure controls, and epitope-masked structural evaluation.

Graduate Research Assistant **07/2024–07/2026**
Department of Food Science and Technology, University of Nebraska–Lincoln, Lincoln, NE
- Coordinated a 12-week randomized crossover clinical trial (NCT06935435); assisted a biobank study.
- Co-developed trial protocol and consent form; authored 4 SOPs; ran 200+ clinic visits; collected, processed, and analyzed samples and data.
- Led a systematic review (PROSPERO CRD420261350487): used ML tools to screen 1,800+ records; LLM-assisted data extraction with human verification.

PUBLICATIONS & CERTIFICATES

Published:
- Xiao J, et al. Foods. 2023;12(15):2964 (Co-author)
- Deng X, et al. J Sci Food Agric. 2023;103(9):4257–4267 (Co-author)

Certificates (CITI Program):
GCP for Clinical Trials with Investigational Drugs and Biologics; Biomedical Responsible Conduct of Research; Biomedical Investigators and Key Personnel; Conflicts of Interest

WORK AND LEADERSHIP EXPERIENCE

Mentor & Co-supervisor *University of Nebraska–Lincoln, Lincoln, NE* **2023–2026**
- Mentored 3 students, co-supervised 1 student with PI.

Graduate Representative *University of Nebraska–Lincoln, Lincoln, NE* **2024–2026**
- Organized 3 professional development workshops; advocated for food science graduate students.

Graduate Student Advisory Board Member *University of Nebraska–Lincoln, Lincoln, NE* **2024–2025**
- Founded a peer-mentoring program pairing new graduate students with mentors.

Teaching Assistant (Internship) *Xi'an New Oriental Education & Technology Group, China* **2022–2023**
- Coached students in IELTS exam preparation.

Communications Lead *Northwest A&F University, China* **2020–2022**
- Directed a 13-member team producing weekly newsletters and videos.

SKILLS

Computational: AlphaFold3, ESMFold2, RFantibody, BindCraft2, RFD4; AI coding agents (Claude Code, Codex)
Clinical Research: trial design and coordination, IRB/GCP compliance, grant writing
Wet Lab: PCR, human blood and stool processing, in vitro fermentation, spectrophotometric assays
Statistics & Languages: multivariate and regression interpretation; Mandarin (native), English (fluent)